

RUI SOUSA

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EDUCATION

Northwestern University, Evanston, IL

2021-2027 (Expected)

PhD Student, Economics Department

- Master of Arts, 2022

- Relevant coursework: Graduate Econometrics, Macro, Micro, Asset Pricing & Corporate Finance (Kellogg Business School)

Nova School of Business and Economics, Lisbon, Portugal

2016-2019

MSc Economics, GPA 19/20 (A) - Novo Banco Master's Scholarship

- Exchange Program at Copenhagen University

University of Porto, Porto, Portugal

2013-2016

BSc Economics, GPA: 18/20 (A)

RESEARCH ACTIVITY

The Aggregate Green Elasticity of Substitution (**Draft**)

- Using a novel semi-structural model, I provide the first quantification of the economy-wide elasticity of substitution between clean and polluting energy sources — a key parameter for analyzing the macro-dynamics of climate change and the green transition
- After deriving a new equilibrium relationship between energy consumption and observable prices, I exploit U.S. states' heterogeneous energy mixes to implement a shift-share instrumental variables strategy, estimating an aggregate elasticity of 0.5
- I construct an analytical mapping between sectoral elasticities and their aggregate counterpart. Using 2022 data to calibrate the model, I find that energy end-users electrify more easily than the electricity sector shifts to green sources, recovering elasticities of 0.8 and 0.52, respectively

Selected Work in Progress

- Climate Change and the Sovereign Debt Overhang
 - I document the increased frequency and intensity of extreme climatic events for different carbon pathways, map their impact on GDP, and calibrate a sovereign default model with disasters to study the interlinkage between public debt and climate change in EMEs
- The Distributional Effects of Climate Change
 - I use local projections with high-dimensional fixed effects on a panel of countries to estimate the effect of climate change on the income distribution

PROFESSIONAL EXPERIENCE

London School of Economics, London, United Kingdom

Sep 2020 - Sep 2021

Research Assistant, Centre For Macroeconomics. Advisors: Profs. Benjamin Moll & Ricardo Reis

- Solved continuous-time heterogeneous agent models numerically using finite-difference methods
- Modeled rare inflation events implementing threshold selection algorithms and maximum likelihood estimation
- Analyzed consumer and professional inflation expectations microdata

European Central Bank, Frankfurt Am Main, Germany

Jun 2018 - Sep 2021

Analyst, Crisis Management - Quantitative Cluster

- Contributed to early warning models processing 1M+ daily observations from global media database and implementing natural language processing
- Analyzed liquidity provisions during stress events and the impact of banking policy on profitability
- Contributed to the impact assessment of Basel III on MREL requirements

Trainee, Stress Test Modelling

- Implemented causal inference methods (regression discontinuity, triple difference-in-differences) on large supervisory banking panel datasets to identify the impact of stress-tests and capital requirements on risk-taking and profitability
- Contributed to credit risk assessment tool using historical loan portfolio data

TOOLS & SKILLS

- **Programming:** R, Python, Matlab
- **Computing:** SLURM (HPC), Git, $\text{\LaTeX}$
- **Datasets:** Compustat, CES, PSID, MER, SEDS, SPF, SCE, CMIP6, Census
- **Languages:** English, Portuguese (fluent); French, Spanish (conversational)
- **Time-series:** State-space & non-linear models, time-varying volatility, bayesian inference, forecasting
- **Microeconometrics:** Shift-share IV, DiD, RDD, statistical learning
- **Modelling:** Dynare, sequence-space jacobian methods